Classical Electron

in

Retarded Minkowski Space

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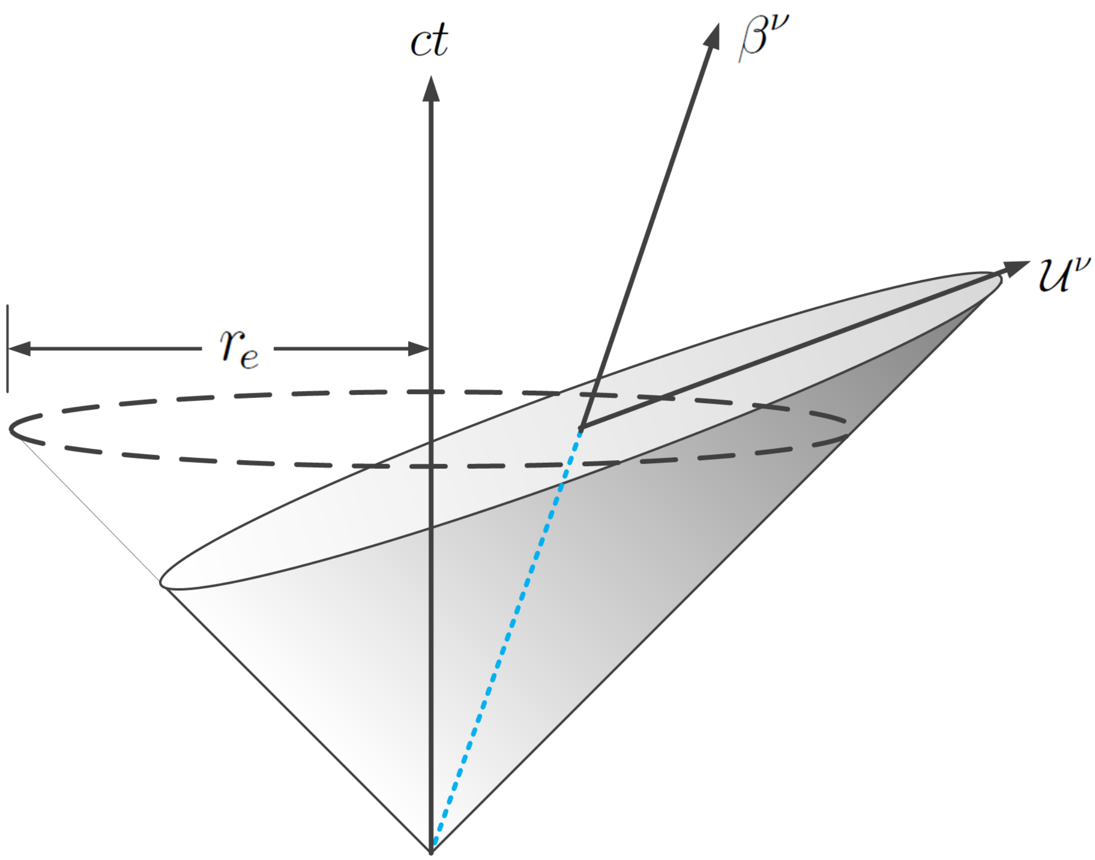

Abstract

Minkowski space is cast into a spherically based four coordinate system derived from principles related to the theory of retarded potentials. The spherically-based system is highly efficient and allows for straight-forward construction of scalars, vectors, and gauge invariant tensors associated with the conventional Maxwell-Lorentz theory.

Vacuum
Gauge
Electrodynamics

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1 Spherical Four-Coordinates

In special relativity, the set of Cartesian unit vectors (labeled $\vec{\mathrm{e}}_\mu$) produce the flat metric tensor via $\mathrm{g}_{\mu\nu} = \vec{\mathrm{e}}_\mu \cdot \vec{\mathrm{e}}_\nu$. Two arbitrary vectors $\vec{\mathrm{A}} = \mathrm{A}^\nu \vec{\mathrm{e}}_\nu$ and $\vec{\mathrm{B}} = \mathrm{B}^\nu \vec{\mathrm{e}}_\nu$ will then combine to form the inner product

$$\vec{\mathrm{A}} \cdot \vec{\mathrm{B}} = \mathrm{A}^\mu \vec{\mathrm{e}}_\mu \cdot \mathrm{B}^\nu \vec{\mathrm{e}}_\nu = \mathrm{A}^\mu \mathrm{B}^\nu \vec{\mathrm{e}}_\mu \cdot \vec{\mathrm{e}}_\nu = \mathrm{A}^\mu \mathrm{B}^\nu \mathrm{g}_{\mu\nu} = \mathrm{A}_\nu \mathrm{B}^\nu \tag{1.1}$$

While this is an example of the proper use of unit vectors, physicists don't always use them[1] for calculations in special relativity. But this formalism becomes important for the introduction of the spherical based four coordinate system where—in addition to the Cartesian basis—two additional sets of unit basis vectors are required.

1.1 A Spherical Basis for Minkowski Space

An inertial electron moves through flat spacetime. At the retarded time ct_r, the particles' position along its worldline is $\vec{r_s} = \vec{r_s}(ct_r)$. As in the spacetime diagram of Figure 1, the null vector $\vec{R}$ traces the motion of a light ray from the source to an arbitrary present time point $\vec{r}$, and satisfies the simple relation

$$\vec{R} \equiv \vec{r} - \vec{r_s} \tag{1.2}$$

The null vector can also be expanded in terms of components along timelike and spacelike unit four vectors $\vec{\beta}$ and $\vec{U}$ by writing

$$\vec{R} = (\vec{R} \cdot \vec{\beta})\vec{\beta} - (\vec{R} \cdot \vec{U})\,\vec{U} = \rho(\vec{\beta} + \vec{U}) \tag{1.3}$$

The last equality defines the Lorentz scalar

$$\rho \equiv \vec{R} \cdot \vec{\beta} = R^\nu \beta_\nu \qquad \text{or} \qquad \rho \equiv -\vec{R} \cdot \vec{U} = -R^\nu U_\nu \tag{1.4}$$

Orthogonal Unit Four Vectors: The orthogonality condition $\vec{\beta} \cdot \vec{U} = 0$ can be verified by writing out the components of each vector, but it also follows directly from the light cone condition $\vec{R} \cdot \vec{R} = 0$ since the norms of $\vec{\beta}$ and $\vec{U}$ cancel. An interesting possibility is the construction of two additional covariant four vectors $(\vec{\vartheta}, \vec{\varphi})$ orthogonal to each other and also to $\vec{\beta}$ and $\vec{U}$. If the new four-vectors can be found, this would define a basis for Minkowski space and generate components of the flat spacetime metric

$$\mathrm{g}^{\mu\nu} = \begin{bmatrix} \vec{\beta} \cdot \vec{\beta} & \vec{\beta} \cdot \vec{U} & \vec{\beta} \cdot \vec{\vartheta} & \vec{\beta} \cdot \vec{\varphi} \\ \vec{U} \cdot \vec{\beta} & \vec{U} \cdot \vec{U} & \vec{U} \cdot \vec{\vartheta} & \vec{U} \cdot \vec{\varphi} \\ \vec{\vartheta} \cdot \vec{\beta} & \vec{\vartheta} \cdot \vec{U} & \vec{\vartheta} \cdot \vec{\vartheta} & \vec{\vartheta} \cdot \vec{\varphi} \\ \vec{\varphi} \cdot \vec{\beta} & \vec{\varphi} \cdot \vec{U} & \vec{\varphi} \cdot \vec{\vartheta} & \vec{\varphi} \cdot \vec{\varphi} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} \tag{1.5}$$

[1]Several chapters in Jackson's *Classical Electrodynamics* incorrectly refer to a symbol like V^α as a vector. But V^α more accurately represents the components of a vector.

Collectively, the set of vectors $\{\vec{\beta}, \vec{U}, \vec{\vartheta}, \vec{\varphi}\}$ may be referred to as a ***Spherical Basis*** for ***Retarded Minkowski Space***. While the angular four-vectors $\vec{\vartheta}$ and $\vec{\varphi}$ are currently unknown, they can be derived by first writing the space components of $\vec{R}$ in terms of spherical polar coordinates

$$\boldsymbol{R} = R\hat{\boldsymbol{n}} = (R\sin\theta\cos\phi, R\sin\theta\sin\phi, R\cos\theta) \tag{1.6}$$

Applying derivatives determines:

$$\frac{\partial \boldsymbol{R}}{\partial R} = \hat{\boldsymbol{n}} \qquad \frac{\partial \boldsymbol{R}}{\partial \theta} = R\hat{\boldsymbol{\theta}} \qquad \frac{\partial \boldsymbol{R}}{\partial \phi} = R\sin\theta\hat{\boldsymbol{\phi}} \tag{1.7}$$

Covariant vectors $\vec{\vartheta}$ and $\vec{\varphi}$ now follow from derivatives of $\vec{U}$

$$\frac{\partial \vec{U}}{\partial \theta} = \frac{R}{\rho}\vec{\vartheta} \qquad \frac{\partial \vec{U}}{\partial \phi} = \frac{R\sin\theta}{\rho}\vec{\varphi} \tag{1.8}$$

Details of (1.8) follow from

$$\frac{\partial}{\partial \theta}\left[\frac{\vec{R}}{\rho}\right] = \frac{\partial}{\partial \theta}\left[\frac{(1,\hat{\boldsymbol{n}})}{\gamma(1-\hat{\boldsymbol{n}}\cdot\boldsymbol{\beta})}\right] = \left[\frac{(0,\hat{\boldsymbol{\theta}})}{\gamma(1-\hat{\boldsymbol{n}}\cdot\boldsymbol{\beta})} + \frac{\boldsymbol{\beta}\cdot\hat{\boldsymbol{\theta}}(1,\hat{\boldsymbol{n}})}{\gamma(1-\hat{\boldsymbol{n}}\cdot\boldsymbol{\beta})^2}\right] = \frac{R}{\rho}\vec{\vartheta}$$

$$\frac{\partial}{\partial \phi}\left[\frac{\vec{R}}{\rho}\right] = \frac{\partial}{\partial \phi}\left[\frac{(1,\hat{\boldsymbol{n}})}{\gamma(1-\hat{\boldsymbol{n}}\cdot\boldsymbol{\beta})}\right] = \sin\theta\left[\frac{(0,\hat{\boldsymbol{\phi}})}{\gamma(1-\hat{\boldsymbol{n}}\cdot\boldsymbol{\beta})} + \frac{\boldsymbol{\beta}\cdot\hat{\boldsymbol{\phi}}(1,\hat{\boldsymbol{n}})}{\gamma(1-\hat{\boldsymbol{n}}\cdot\boldsymbol{\beta})^2}\right] = \frac{R}{\rho}\sin\theta\,\vec{\varphi}$$

An orthonormal set of unit four-vectors with raised indices—written explicitly as functions of the accompanying set of three-vectors $(\hat{\boldsymbol{n}}, \hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\phi}})$ —may be summarized by writing

$$\vec{\beta} = \beta^\nu \vec{\mathrm{e}}_\nu = \begin{bmatrix} \gamma \\ \gamma\boldsymbol{\beta} \end{bmatrix} \qquad \beta^\nu\beta_\nu = 1 \tag{1.10a}$$

$$\vec{U} = U^\nu \vec{\mathrm{e}}_\nu = \frac{1}{\gamma(1-\boldsymbol{\beta}\cdot\hat{\boldsymbol{n}})}\begin{bmatrix} 1 \\ \hat{\boldsymbol{n}} \end{bmatrix} - \begin{bmatrix} \gamma \\ \gamma\boldsymbol{\beta} \end{bmatrix} \qquad U^\nu U_\nu = -1 \tag{1.10b}$$

$$\vec{\vartheta} = \vartheta^\nu \vec{\mathrm{e}}_\nu = \frac{1}{(1-\boldsymbol{\beta}\cdot\hat{\boldsymbol{n}})}\begin{bmatrix} \boldsymbol{\beta}\cdot\hat{\boldsymbol{\theta}} \\ \hat{\boldsymbol{\theta}} + \boldsymbol{\beta}\times\hat{\boldsymbol{\phi}} \end{bmatrix} \qquad \vartheta^\nu\vartheta_\nu = -1 \tag{1.10c}$$

$$\vec{\varphi} = \varphi^\nu \vec{\mathrm{e}}_\nu = \frac{1}{(1-\boldsymbol{\beta}\cdot\hat{\boldsymbol{n}})}\begin{bmatrix} \boldsymbol{\beta}\cdot\hat{\boldsymbol{\phi}} \\ \hat{\boldsymbol{\phi}} - \boldsymbol{\beta}\times\hat{\boldsymbol{\theta}} \end{bmatrix} \qquad \varphi^\nu\varphi_\nu = -1 \tag{1.10d}$$

This set represents a very powerful basis for Minkowski space, capable of sampling the electromagnetic fields of an electron—even if the particle is accelerating. Essentially, the entire theory of the classical electron can be cast into this basis, but it is not an overstatement to point out that retarded Minkowski space can be applied to any theory of physics designed to be consistent with special relativity.

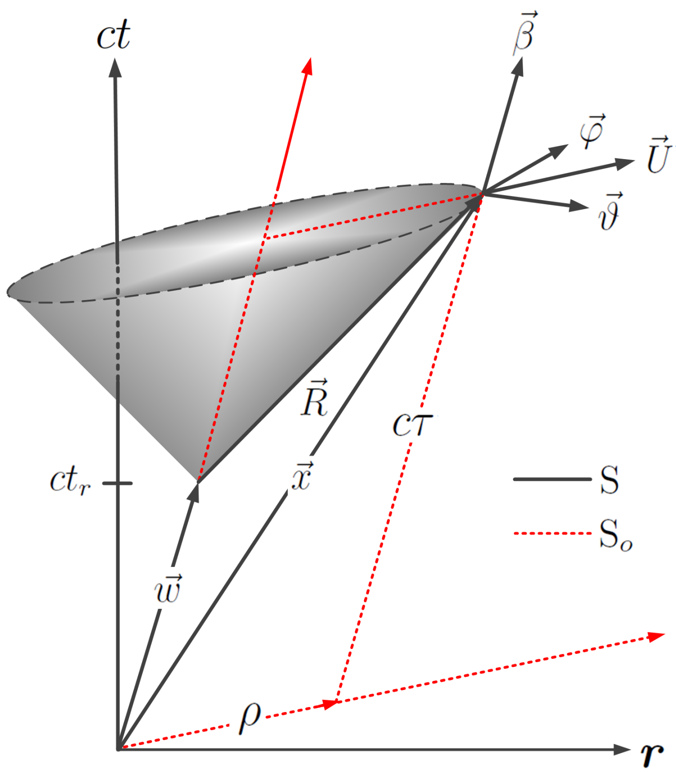

Figure 1: *Orthonormal set of unit four-vectors at the point $\vec{x}$ defined relative to the retarded position of an electron moving along its world line.*

1.2 Coordinate Transformation

Basis vectors summarized in equation (1.10) can be placed on a more solid theoretical foundation by deriving them from a coordinate transformation

$$(ct, x, y, z) = f(c\tau, \rho, \theta, \phi) \tag{1.11}$$

where $c\tau$ and ρ are Lorentz scalars and all coordinates are assumed to be independent of each other. The explicit form of the transformation results by simply projecting $\vec{r}$ onto the spherical basis:

$$\vec{r} = (\vec{r} \cdot \vec{\beta})\vec{\beta} - (\vec{r} \cdot \vec{U})\,\vec{U} \tag{1.12}$$

Coefficients can be calculated rigorously or they can be read directly from Figure 1 —leading the vector relation

$$\boxed{\vec{r} = c\tau\vec{\beta} + \rho\vec{U}} \tag{1.13}$$

Derivatives with respect to the new coordinates will generate the basis vectors $\vec{\mathcal{S}}_\nu$:

$$\vec{\mathcal{S}}_\tau = \frac{\partial \vec{x}}{\partial c\tau} = \vec{\beta} \qquad \vec{\mathcal{S}}_\rho = \frac{\partial \vec{x}}{\partial \rho} = \vec{U} \tag{1.14a}$$

$$\vec{\mathcal{S}}_\theta = \frac{\partial \vec{x}}{\partial \theta} = R\,\vec{\vartheta} \qquad \vec{\mathcal{S}}_\phi = \frac{\partial \vec{x}}{\partial \phi} = R\sin\theta\,\vec{\varphi} \tag{1.14b}$$

The scale factors can be read directly from the basis vectors

$$(h_\tau, h_\rho, h_\theta, h_\phi) = (1, 1, R, R\sin\theta) \tag{1.15}$$

while dot products of the basis vectors determine the metric tensor[2],

$$\vec{\mathcal{S}}_\alpha \cdot \vec{\mathcal{S}}_\beta \equiv \mathrm{g}^{[sph]}_{\alpha\beta} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -R^2 & 0 \\ 0 & 0 & 0 & -R^2\sin^2\theta \end{bmatrix} \tag{1.16}$$

Components of the metric also follow by calculation of the line element ds^2 —derivable from the differential of $\vec{r}$ in equation (1.13):

$$\begin{aligned} d\vec{x} &= dc\tau\vec{\beta} + d\rho\vec{U} + \rho\frac{\partial\vec{U}}{\partial\theta}d\theta + \rho\frac{\partial\vec{U}}{\partial\phi}d\phi \\ &= \vec{\beta}\,dc\tau + \vec{U}\,d\rho + R\vec{\vartheta}\,d\theta + R\sin\theta\vec{\varphi}\,d\phi \end{aligned} \tag{1.17}$$

The line element is

$$ds^2 = d\vec{x}\cdot d\vec{x} = dx^\nu dx_\nu = c^2d\tau^2 - d\rho^2 - R^2d\theta^2 - R^2\sin^2\theta d\phi^2 \tag{1.18}$$

and may also be written

$$ds^2 = c^2d\tau^2\left[1 - \left(\frac{d\rho}{dc\tau}\right)^2 - \left(\frac{d[R\theta]}{dc\tau}\right)^2 - \left(\frac{d[R\sin\theta\phi]}{dc\tau}\right)^2\right] \tag{1.19}$$

But if $c\tau$ represents proper time, invariance of the line element requires $ds^2 = c^2d\tau^2$. This means that all derivatives inside the braces on the right side of equation (1.19) are zero. This highlights the essential property of the spherical basis where the four-velocity $\vec{\beta}$ is contained within the basis vectors themselves and fails to exist in the line element.

[2]The metric can be evaluated at classical radius $R = r_e$ or some other radius. This is sphere about the particle with all points on the sphere occuring at the same time. But the instantaneous position of the particle is not at the center of the sphere.

Inverse Transformation: Like many coordinate transformations in physics, equation (1.13) can be inverted so that each of the coordinates $\{c\tau, \rho, \theta, \phi\}$ are functions of the Cartesian coordinates $\{ct, x, y, z\}$. A representation of the inverse transformation is

$$c\tau = ct_r/\gamma + \rho \qquad\qquad \rho = \gamma R - \gamma \boldsymbol{R}\cdot\boldsymbol{\beta}$$

$$\theta = \tan^{-1}\left[\frac{(R_x^2 + R_y^2)^{1/2}}{R_z}\right] \qquad\qquad \phi = \tan^{-1}\left[\frac{R_y}{R_x}\right] \tag{1.20a}$$

Unfortunately, these formulas are somewhat problematic since the Cartesian coordinates are buried within ct_r and the components of $\boldsymbol{R}$. Nevertheless, each spherical coordinate can be set to a constant value which will define a three-dimensional hypersurface in Minkowski space. These surfaces may be difficult to visualize except in the rest frame where the spacelike surfaces become identical to those of the standard spherical-polar coordinate system. Dual basis vectors (one-forms), similar to those in equation (1.14) except with raised indices, follow from the inverse transformation as:

$$\vec{\mathcal{J}}^{\,\tau} = \partial_\nu c\tau\, \vec{\omega}^\nu = \beta_\nu\, \vec{\omega}^\nu \qquad\qquad \vec{\mathcal{J}}^{\,\rho} = \partial_\nu \rho\, \vec{\omega}^\nu = -\,U_\nu\, \vec{\omega}^\nu \tag{1.21a}$$

$$\vec{\mathcal{J}}^{\,\theta} = \partial_\nu \theta\, \vec{\omega}^\nu = -\frac{1}{R}\vartheta_\nu\, \vec{\omega}^\nu \qquad\qquad \vec{\mathcal{J}}^{\,\phi} = \partial_\nu \phi\, \vec{\omega}^\nu = -\frac{1}{R\sin\theta}\varphi_\nu\, \vec{\omega}^\nu \tag{1.21b}$$

Unlike (1.14), the spacelike vectors derived from the inverse transformation appear with minus signs and scale factors in the denominators. These properties determine the dot product between vectors and one-forms

$$\vec{\mathcal{S}}_\mu \bullet \vec{\mathcal{J}}^{\,\nu} = \vec{\mathrm{e}}_\mu \cdot \vec{\omega}^{\,\nu} = \delta_\mu{}^\nu \tag{1.22}$$

Christoffel Symbols: Derivatives of the basis vectors define ten independent Christoffel symbols thru the relations

$$\frac{\partial \vec{\mathcal{S}}_\mu}{\partial q^\nu} = \Gamma^\lambda_{\mu\nu}\, \vec{\mathcal{S}}_\lambda \qquad\qquad \frac{\partial \vec{\mathcal{J}}^{\,\mu}}{\partial q^\nu} = -\Gamma^\mu_{\lambda\nu}\, \vec{\mathcal{J}}^{\,\lambda} \tag{1.23}$$

but they also follow directly from the metric tensor:

$$\Gamma^\alpha_{\mu\nu} = \tfrac{1}{2}\mathrm{g}^{\lambda\alpha}\left[\mathrm{g}_{\lambda\mu,\nu} + \mathrm{g}_{\lambda\nu,\mu} - \mathrm{g}_{\mu\nu,\lambda}\right] \tag{1.24}$$

Christoffel symbols for the spherical basis are summarized in Figure 2 and may be used to verify that all components of the curvature tensor are zero. For $\beta \to 0$, several of the symbols vanish while the remaining non-zero symbols reduce to those of a 3D spherical-polar coordinate system.

1. $\Gamma^{\phi}_{\rho\phi} = \dfrac{1}{\rho}$
2. $\Gamma^{\theta}_{\rho\theta} = \dfrac{1}{\rho}$
3. $\Gamma^{\rho}_{\theta\theta} = -\dfrac{\rho}{\gamma^2(1-\hat{\boldsymbol{n}}\cdot\boldsymbol{\beta})^2}$
4. $\Gamma^{\rho}_{\phi\phi} = -\dfrac{\rho\sin^2\theta}{\gamma^2(1-\hat{\boldsymbol{n}}\cdot\boldsymbol{\beta})^2}$
5. $\Gamma^{\theta}_{\theta\theta} = \dfrac{\boldsymbol{\beta}\cdot\hat{\boldsymbol{\theta}}}{1-\hat{\boldsymbol{n}}\cdot\boldsymbol{\beta}}$
6. $\Gamma^{\phi}_{\theta\theta} = -\dfrac{\boldsymbol{\beta}\cdot\hat{\boldsymbol{\phi}}}{1-\hat{\boldsymbol{n}}\cdot\boldsymbol{\beta}}$
7. $\Gamma^{\phi}_{\theta\phi} = \cot\theta + \dfrac{\boldsymbol{\beta}\cdot\hat{\boldsymbol{\theta}}}{1-\hat{\boldsymbol{n}}\cdot\boldsymbol{\beta}}$
8. $\Gamma^{\theta}_{\theta\phi} = \dfrac{\boldsymbol{\beta}\cdot\hat{\boldsymbol{\phi}}}{1-\hat{\boldsymbol{n}}\cdot\boldsymbol{\beta}}\sin\theta$
9. $\Gamma^{\theta}_{\phi\phi} = -\sin\theta\cos\theta - \dfrac{\boldsymbol{\beta}\cdot\hat{\boldsymbol{\theta}}}{1-\hat{\boldsymbol{n}}\cdot\boldsymbol{\beta}}\sin^2\theta$
10. $\Gamma^{\phi}_{\phi\phi} = \dfrac{\boldsymbol{\beta}\cdot\hat{\boldsymbol{\phi}}}{1-\hat{\boldsymbol{n}}\cdot\boldsymbol{\beta}}\sin\theta$

Figure 2: *Ten independent Christoffel symbols in the spherical basis.*

1.3 Differential Operations on Scalars and Vectors

In this section, four-gradient, four-divergence, and the wave operation are applied in the spherical basis, followed by several examples for each. First however, it is important to project an arbitrary vector (or vector field) $\vec{V}$ onto the spherical basis. This can be accomplished with either raised or lowered indices:

$$\vec{V} = V^\nu \vec{\mathcal{S}}_\nu \qquad \text{or} \qquad \vec{V} = V_\nu \vec{\mathcal{J}}^\nu \tag{1.25}$$

Choosing lowered indices one may write

$$\vec{V} = \sum_\nu (\vec{V}\cdot\vec{\mathcal{J}}^\nu)\vec{\mathcal{S}}_\nu = (\vec{V}\cdot\vec{\mathcal{J}}^\tau)\vec{\mathcal{S}}_\tau + (\vec{V}\cdot\vec{\mathcal{J}}^\rho)\vec{\mathcal{S}}_\rho + (\vec{V}\cdot\vec{\mathcal{J}}^\theta)\vec{\mathcal{S}}_\theta + (\vec{V}\cdot\vec{\mathcal{J}}^\phi)\vec{\mathcal{S}}_\phi \tag{1.26}$$

But the dot products are easy to evaluate. The first term is

$$\vec{V}\cdot\vec{\mathcal{J}}^\tau = V^\nu \vec{\mathrm{e}}_\nu \cdot \beta_\alpha \vec{\omega}^\alpha = V^\nu \beta_\alpha \cdot \delta^\alpha_{\ \nu} = V^\alpha \beta_\alpha \tag{1.27}$$

Inserting all dot products and cancelling unit vectors from both sides of the resulting equation generates an expansion for the components of $\vec{V}$

$$V^\nu = (V^\alpha\beta_\alpha)\beta^\nu - (V^\alpha U_\alpha)U^\nu - (V^\alpha\vartheta_\alpha)\vartheta^\nu - (V^\alpha\varphi_\alpha)\varphi^\nu \tag{1.28}$$

An example relating to (1.28) follows by considering the set of vacuum gauge velocity potentials $\vec{A} = A^\nu \vec{\mathrm{e}}_\nu$ in the spherical basis. This is a radial vector from the retarded position of the electron so components along ϑ^ν and φ^ν are automatically zero. An ordered 4-tuple in the spherical basis follows by writing

$$A^\nu \longrightarrow \left[\frac{e}{\rho},\, \frac{e}{\rho},\, 0,\, 0\right] \tag{1.29}$$

1.3.1 Four-Gradient

In the spherical basis an arbitrary scalar field $\psi = \psi(c\tau, \rho, \theta, \phi)$ has a four-gradient which can be written

$$\vec{\nabla}\psi = \frac{\partial\psi}{\partial c\tau}\vec{\jmath}^{\,\tau} + \frac{\partial\psi}{\partial\rho}\vec{\jmath}^{\,\rho} + \frac{\partial\psi}{\partial\theta}\vec{\jmath}^{\,\theta} + \frac{\partial\psi}{\partial\phi}\vec{\jmath}^{\,\phi} = \partial_\nu\psi\,\vec{\omega}^\nu \tag{1.30}$$

The basis vectors on the right are determined in equation (1.21a) and they can be immediately inserted above. Now eliminate the unit vector $\vec{\omega}^\nu$ from both sides of the equation by forming the dot product $\vec{\nabla}\psi \cdot \vec{e}_\nu$. What remains is

$$\partial_\nu\psi = \frac{\partial\psi}{\partial c\tau}\beta_\nu - \frac{\partial\psi}{\partial\rho}U_\nu - \frac{1}{R}\frac{\partial\psi}{\partial\theta}\vartheta_\nu - \frac{1}{R\sin\theta}\frac{\partial\psi}{\partial\phi}\varphi_\nu \tag{1.31}$$

Proper use of this operator equation can be deceptive, especially when applying it in situations involving particle accelerations.

Example 1: Complex Exponential: The complex exponential is $\mathrm{e}^{ik^\sigma x_\sigma}$ where k^σ is an arbitrary constant wave vector. To understand the argument in the spherical basis, write x_σ in terms of transformation equation (1.13) and write the inner product:

$$k^\sigma x_\sigma = c\tau k^\sigma\beta_\sigma + \rho k^\sigma U_\sigma \tag{1.32}$$

Now apply the gradient operator

$$\partial_\nu[k^\sigma x_\sigma] = \left[\beta_\nu\partial_\tau - U_\nu\partial_\rho - \frac{\vartheta_\nu}{R}\partial_\theta - \frac{\varphi_\nu}{R\sin\theta}\partial_\phi\right] \cdot [c\tau k^\sigma\beta_\sigma + \rho k^\sigma U_\sigma] \tag{1.33}$$

The first term on the right side of (1.32) depends only on $c\tau$ while the second term depends on all three spatial coordinates (ρ, θ, ϕ). Implementing derivatives generates

$$\partial_\nu[k^\sigma x_\sigma] = \left[(k^\sigma\beta_\sigma)\beta_\nu - (k^\sigma U_\sigma)U_\nu - \frac{\rho}{R}(k^\sigma\partial_\theta U_\sigma)\vartheta_\nu - \frac{\rho}{R\sin\theta}(k^\sigma\partial_\phi U_\sigma)\varphi_\nu\right] \tag{1.34}$$

Now appeal to equation (1.28) to simplify

$$\partial_\nu[k^\sigma x_\sigma] = (k^\sigma\beta_\sigma)\beta_\nu - (k^\sigma U_\sigma)U_\nu - (k^\sigma\vartheta_\sigma)\vartheta_\nu - (k^\sigma\varphi_\sigma)\varphi_\nu = k^\nu \tag{1.35}$$

This implies the well known relation

$$\partial_\nu \mathrm{e}^{ik^\sigma x_\sigma} = ik_\nu \mathrm{e}^{ik^\sigma x_\sigma}, \tag{1.36}$$

but now it's been proved in the spherical basis.

Example 2: Dirac Particle: Equation (1.31) can be used to cast Dirac electron theory into the spherical basis. The well-known free particle Dirac equation is $[i\hbar c\gamma^\nu\partial_\nu - mc^2]\,\psi = 0$. Inserting ∂_ν in the new basis determines

$$i\hbar c\left[\gamma^\nu\beta_\nu\partial_\tau - \gamma^\nu U_\nu\partial_\rho - \frac{1}{R}\gamma^\nu\vartheta_\nu\partial_\theta - \frac{1}{R\sin\theta}\gamma^\nu\varphi_\nu\partial_\phi - mc^2\right]\psi = 0 \tag{1.37}$$

This equation may also be written more compactly as

$$i\hbar c\left[\gamma^\tau\partial_\tau + \gamma^\rho\partial_\rho + \frac{1}{R}\gamma^\theta\partial_\theta + \frac{1}{R\sin\theta}\gamma^\phi\partial_\phi - mc^2\right]\psi = 0 \tag{1.38}$$

where the new gamma matrices, collected together as a set, are

$$\gamma^\nu \equiv [\gamma^\tau, \gamma^\rho, \gamma^\theta, \gamma^\phi] = [\gamma^\nu\beta_\nu,\ -\gamma^\nu U_\nu,\ -\gamma^\nu\vartheta_\nu,\ -\gamma^\nu\varphi_\nu] \tag{1.39}$$

These matrices satisfy $[\gamma^\tau]^2 = 1$ and $[\gamma^\rho]^2 = [\gamma^\theta]^2 = [\gamma^\phi]^2 = -1$ similar to gamma matrices in the Cartesian theory. Evaluating all possible anti-commutation relations determines components of the flat spacetime metric tensor from

$$\{\gamma^\mu, \gamma^\nu\} = 2\mathrm{I}_4 \cdot \mathrm{g}^{\mu\nu} \tag{1.40}$$

This equation requires the individual elements γ^ν to be viewed as generators of the Clifford algebra $C\ell_{1,3}(C)$, except determined in the spherical basis. The validity of this assertion is confirmed by construction $\gamma^5 = i\gamma^\tau\gamma^\rho\gamma^\theta\gamma^\phi$.

1.3.2 Four-Divergence

For an arbitrary vector $\vec{V}$ in four-dimensional spacetime, a general formula for the four-divergence in terms of scale factors is

$$\partial_\nu V^\nu = \frac{1}{h_\tau h_\rho h_\theta h_\phi}\left[\frac{\partial}{\partial c\tau}(h_\rho h_\theta h_\phi V^\tau) + \frac{\partial}{\partial\rho}(h_\tau h_\theta h_\phi V^\rho)\right] + \frac{1}{h_\tau h_\rho h_\theta h_\phi}\left[\frac{\partial}{\partial\theta}(h_\tau h_\rho h_\phi V^\theta) + \frac{\partial}{\partial\phi}(h_\tau h_\rho h_\theta V^\phi)\right] \tag{1.41}$$

Inserting scale factors from equation (1.15) determines

$$\partial_\nu V^\nu = \frac{\partial V^\tau}{\partial c\tau} + \frac{1}{R^2}\frac{\partial}{\partial\rho}(R^2V^\rho) + \frac{1}{R^2\sin\theta}\frac{\partial}{\partial\theta}(R\sin\theta V^\theta) + \frac{1}{R^2\sin\theta}\frac{\partial}{\partial\phi}(RV^\phi) \tag{1.42}$$

This is a clean representation of the four-divergence, but an application of this formula to an arbitrary four-vector may require knowledge of derivatives of R with respect to the spacelike variables:

$$\frac{\partial R}{\partial\rho} = \frac{R}{\rho} \qquad \frac{\partial R}{\partial\theta} = \frac{\gamma R^2}{\rho}\boldsymbol{\beta}\cdot\hat{\boldsymbol{\theta}} \qquad \frac{\partial R}{\partial\phi} = \frac{\gamma R^2}{\rho}\sin\theta\boldsymbol{\beta}\cdot\hat{\boldsymbol{\phi}} \tag{1.43}$$

For the limit when the four-velocity tends to zero it is easy to reconcile the spacelike terms in (1.42) with the well known three-space divergence. As a simple example, the four-divergence can be applied to Liénard-Wiechert potentials $\vec{A} = e\vec{\beta}/\rho$ —having a single component along the timelike direction. By inspection, $\vec{\nabla}\cdot\vec{A} = 0$.

1.3.3 Wave Operator

According to the theory of curvilinear coordinate systems, the wave operator applied to a scalar ψ will be determined from

$$\Box^2\psi = \frac{1}{h_\tau h_\rho h_\theta h_\phi}\left[\frac{\partial}{\partial c\tau}\left(\frac{h_\rho h_\theta h_\phi}{h_\tau}\frac{\partial\psi}{\partial c\tau}\right) - \frac{\partial}{\partial\rho}\left(\frac{h_\tau h_\theta h_\phi}{h_\rho}\frac{\partial\psi}{\partial\rho}\right)\right]$$
$$- \frac{1}{h_\tau h_\rho h_\theta h_\phi}\left[\frac{\partial}{\partial\theta}\left(\frac{h_\tau h_\rho h_\phi}{h_\theta}\frac{\partial\psi}{\partial\theta}\right) - \frac{\partial}{\partial\phi}\left(\frac{h_\tau h_\rho h_\theta}{h_\phi}\frac{\partial\psi}{\partial\phi}\right)\right] \tag{1.44}$$

Inserting the scale factors from equation (1.15) renders the operator equation

$$\Box^2\psi = \frac{\partial^2\psi}{\partial c\tau^2} - \frac{\partial^2\psi}{\partial\rho^2} - \frac{2}{\rho}\frac{\partial\psi}{\partial\rho} - \frac{1}{R^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\psi}{\partial\theta}\right) - \frac{1}{R^2\sin^2\theta}\cdot\frac{\partial^2\psi}{\partial\phi^2} \tag{1.45}$$

Example 3: The homogeneous wave equation for a scalar field is $\Box^2\psi = 0$. Following Jackson (chapter 16), solutions to this equation may be written

$$\psi(c\tau,\rho,\theta,\phi) = \int_{-\infty}^{\infty}\psi(\omega,\rho,\theta,\phi)\mathrm{e}^{-i\omega\tau}d\omega \tag{1.46}$$

Inserting into the wave equation determines the spherically based Helmholtz equation with solutions

$$\psi(\omega,\rho,\theta,\phi) = \sum_{\ell,m} f_\ell(\rho)\cdot Y_{\ell,m}(\theta,\phi) \tag{1.47}$$

where $Y_{\ell,m}(\theta,\phi)$ are the spherical harmonics and $f_\ell(\rho)$ satisfy the radial equation

$$\left[\frac{\partial^2}{\partial\rho^2} + \frac{2}{\rho}\frac{\partial}{\partial\rho} + k^2 - \frac{\ell(\ell+1)}{R^2}\right] f_\ell(\rho) \tag{1.48}$$

Solutions to this equation are the spherical Bessel, Neumann, and Hankel functions.

Example 4: The theory of retarded potentials requires the electrons' four-velocity to be evaluated at the retarded time $\vec{\beta}(ct_r)$. In the spherically based coordinate system, this requires the four-velocity to be a function of two coordinates $\vec{\beta} = \vec{\beta}(c\tau - \rho)$. The acceleration four vector follows as

$$\vec{a} = \frac{\partial\vec{\beta}}{\partial c\tau} = -\frac{\partial\vec{\beta}}{\partial\rho} \tag{1.49}$$

Now supppose the wave operator is applied to the Liénard-Wiechert potentials:

$$\vec{A} = \frac{e}{\rho}\cdot\vec{\beta}(c\tau - \rho) \tag{1.50}$$

In regions of space away from the source, derivatives are

$$\frac{\partial^2 \vec{A}}{\partial c\tau^2} = \frac{e}{\rho} \cdot \frac{\partial \vec{a}}{\partial c\tau} \qquad \frac{\partial \vec{A}}{\partial \rho} = -\frac{\vec{A}}{\rho} - \frac{e\vec{a}}{\rho} \qquad \frac{\partial^2 \vec{A}}{\partial \rho^2} = \frac{2\vec{A}}{\rho^2} + \frac{2e\vec{a}}{\rho^2} - \frac{e}{\rho} \cdot \frac{\partial \vec{a}}{\partial \rho} \tag{1.51}$$

Combining and canceling terms verifies that Liénard-Wiechert potentials satisfy the vector wave equation

$$\Box^2 \vec{A} = \frac{\partial^2 \vec{A}}{\partial c\tau^2} - \frac{\partial^2 \vec{A}}{\partial \rho^2} - \frac{2}{\rho}\frac{\partial \vec{A}}{\partial \rho} = 0 \tag{1.52}$$

This derivation is perfectly general—covering all possible motions of the electron.

1.4 Three Sets of Unit Basis Vectors

The unit basis vectors in equation (1.10) are generally assumed to be written in the Cartesian basis, but they can also be viewed as functions of the unit three-vectors $(\hat{\boldsymbol{n}}, \hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\phi}})$. Combining these three with the timelike vector $\hat{\boldsymbol{t}} \Leftrightarrow \vec{\mathrm{e}}_o$ defines another basis for Minkowski space—referred to as the ***Radiation Basis***[3]. The name seems appropriate since the unit vector $\hat{\boldsymbol{n}}$ generally describes the direction of an emitted light speed particle. To summarize, there exists a triad of associated elemental sets of unit basis vectors associated with (1.10):

$$\begin{aligned} &\mathrm{S} : \longrightarrow \{\vec{\beta}, \vec{U}, \vec{\vartheta}, \vec{\varphi}\} && \text{Spherical Basis} \\ &\mathrm{C} : \longrightarrow \{\vec{\mathrm{e}}_o, \vec{\mathrm{e}}_x, \vec{\mathrm{e}}_y, \vec{\mathrm{e}}_z\} && \text{Cartesian Basis} \\ &\hat{\boldsymbol{s}} : \longrightarrow \{\hat{\boldsymbol{t}}, \hat{\boldsymbol{n}}, \hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\phi}}\} && \text{Radiation Basis} \end{aligned} \tag{1.54}$$

An immediate concern is the identification of transformation matrices to determine one set from the other. To derive the Spherical basis from the Cartesian basis write

$$\begin{bmatrix} \vec{\beta} \\ \vec{U} \\ \vec{\vartheta} \\ \vec{\varphi} \end{bmatrix} = \begin{bmatrix} \beta_0 & \beta_x & \beta_y & \beta_z \\ U_0 & U_x & U_y & U_z \\ \vartheta_0 & \vartheta_x & \vartheta_y & \vartheta_z \\ \varphi_0 & \varphi_x & \varphi_y & \varphi_z \end{bmatrix} \cdot \begin{bmatrix} \vec{\mathrm{e}}_o \\ \vec{\mathrm{e}}_x \\ \vec{\mathrm{e}}_y \\ \vec{\mathrm{e}}_z \end{bmatrix} \tag{1.55}$$

[3] Another representation of the Radiation basis follows from the definitions

$$\begin{aligned} &\vec{t} = t^\nu \vec{\mathrm{e}}_\nu \quad \longrightarrow \quad t^\nu = (1, \mathbf{0}) && \vec{n} = n^\nu \vec{\mathrm{e}}_\nu \quad \longrightarrow \quad n^\nu = (0, \hat{\boldsymbol{n}}) \\ &\vec{\theta} = \theta^\nu \vec{\mathrm{e}}_\nu \quad \longrightarrow \quad \theta^\nu = (0, \hat{\boldsymbol{\theta}}) && \vec{\phi} = \phi^\nu \vec{\mathrm{e}}_\nu \quad \longrightarrow \quad \phi^\nu = (0, \hat{\boldsymbol{\phi}}) \end{aligned}$$

Components V^ν of an arbitrary vector are then determined by the expansion

$$V^\nu = (V^\alpha t_\alpha) t^\nu - (V^\alpha n_\alpha) n^\nu - (V^\alpha \theta_\alpha) \theta^\nu - (V^\alpha \phi_\alpha) \phi^\nu$$

To write spherical basis vectors in terms of the radiation basis, its easier to write out the 4-tuples:

$$\vec{\beta} = \gamma\hat{\boldsymbol{t}} + \gamma(\boldsymbol{\beta}\cdot\hat{\boldsymbol{n}})\hat{\boldsymbol{n}} + \gamma(\boldsymbol{\beta}\cdot\hat{\boldsymbol{\theta}})\hat{\boldsymbol{\theta}} + \gamma(\boldsymbol{\beta}\cdot\hat{\boldsymbol{\phi}})\hat{\boldsymbol{\phi}}$$

$$\vec{U} = \left[\frac{1}{\gamma(1-\hat{\boldsymbol{n}}\cdot\boldsymbol{\beta})} - \gamma\right]\hat{\boldsymbol{t}} + \left[\frac{1}{\gamma(1-\hat{\boldsymbol{n}}\cdot\boldsymbol{\beta})} - \gamma\boldsymbol{\beta}\cdot\hat{\boldsymbol{n}}\right]\hat{\boldsymbol{n}} - \gamma(\boldsymbol{\beta}\cdot\hat{\boldsymbol{\theta}})\hat{\boldsymbol{\theta}} - \gamma(\boldsymbol{\beta}\cdot\hat{\boldsymbol{\phi}})\hat{\boldsymbol{\phi}}$$

$$\vec{\vartheta} = \left[\frac{\boldsymbol{\beta}\cdot\hat{\boldsymbol{\theta}}}{1-\hat{\boldsymbol{n}}\cdot\boldsymbol{\beta}}\right]\hat{\boldsymbol{t}} + \left[\frac{\boldsymbol{\beta}\cdot\hat{\boldsymbol{\theta}}}{1-\hat{\boldsymbol{n}}\cdot\boldsymbol{\beta}}\right]\hat{\boldsymbol{n}} + \hat{\boldsymbol{\theta}}$$

$$\vec{\varphi} = \left[\frac{\boldsymbol{\beta}\cdot\hat{\boldsymbol{\phi}}}{1-\hat{\boldsymbol{n}}\cdot\boldsymbol{\beta}}\right]\hat{\boldsymbol{t}} + \left[\frac{\boldsymbol{\beta}\cdot\hat{\boldsymbol{\phi}}}{1-\hat{\boldsymbol{n}}\cdot\boldsymbol{\beta}}\right]\hat{\boldsymbol{n}} + \hat{\boldsymbol{\phi}} \tag{1.56}$$

Equations (1.55) and (1.56) can also be combined to solve for radiation basis vectors in terms of Cartesian unit vectors. This requires calculating the inverse of the matrix (1.55). One determines the matrix equation

$$\vec{s} = \Omega\vec{x} \qquad \text{or} \qquad \begin{bmatrix}\hat{\boldsymbol{t}}\\ \hat{\boldsymbol{n}}\\ \hat{\boldsymbol{\theta}}\\ \hat{\boldsymbol{\phi}}\end{bmatrix} = \begin{bmatrix}1 & 0 & 0 & 0\\ 0 & n_x & n_y & n_z\\ 0 & \theta_x & \theta_y & \theta_z\\ 0 & \phi_x & \phi_y & \phi_z\end{bmatrix}\cdot\begin{bmatrix}\vec{\mathrm{e}}_o\\ \vec{\mathrm{e}}_x\\ \vec{\mathrm{e}}_y\\ \vec{\mathrm{e}}_z\end{bmatrix} \tag{1.57}$$

As a reminder, this is not a rest frame transformation. Instead, $\hat{\boldsymbol{n}}$ is a unit vector pointing from the retarded position of a moving particle, with derivatives proportional to $\hat{\boldsymbol{\theta}}$ and $\hat{\boldsymbol{\phi}}$. Nevertheless, each of the transfromation matrices associated with equations (1.55), (1.56), and (1.57) have determinants +1. In addition, Columns and rows of each matrix are vectors with norms $(+1, -1, -1, -1)$.

1.5 Integration in the Spherical Basis

Scale factors are immediately available from equation (1.16) to construct both volume and surface elements using the general rules set forth for curvi-linear coordinate systems. The four-volume element is

$$d^4\mathcal{V} = \rho^2 d\rho\, d\Omega'\, cd\tau \tag{1.58}$$

where the solid angle differential $d\Omega'$ is defined by the integration

$$\oint d\Omega' = \oint \frac{\sin\theta}{\gamma^2(1-\boldsymbol{\beta}\cdot\hat{\boldsymbol{n}})^2} d\theta d\phi = 4\pi \tag{1.59}$$

Spacelike and timelike hyper-surface elements—written as relations among vectors and relations among components—are

$$d^3\vec{\sigma}_s = [\rho^2 d\rho\, d\Omega']\,\vec{\beta} \qquad\qquad d\sigma_s^\mu = [\rho^2 d\rho\, d\Omega']\,\beta^\mu \tag{1.60a}$$

$$d^3\vec{\sigma}_t = -[\rho^2 c d\tau\, d\Omega']\,\vec{U} \qquad\qquad d\sigma_t^\mu = [\rho^2 c d\tau\, d\Omega']\,U^\mu \tag{1.60b}$$

Equating increments of $d\rho$ and $cd\tau$, the surface-elements in (1.60) can be added together (with a minus sign) leading to the null surface element

$$d\sigma_l^\mu = -[\rho d\rho\, d\Omega']\,R^\mu \tag{1.61}$$

2 Classical Electron Theory in the Spherical Basis

The covariance of classical electron theory implies that all properties of the electron can be re-established in retarded Minkowski space. Of particular importance is a re-assessment of particle acceleration fields which divide naturally into left-handed and right-handed components in the radiation basis. Before any discussion of particle electromagnetic fields however, it will prove useful to review the Liénard power formula.

2.1 Power Radiated by an Accelerating Electron

Components of the acceleration four-vector can be determined in the spherical basis from an expansion of $\vec{a}$ using equation (1.28). With $a_\nu\beta^\nu = 0$,

$$a^\nu = -(a^\alpha U_\alpha)\,U^\nu - (a^\alpha\vartheta_\alpha)\vartheta^\nu - (a^\alpha\varphi_\alpha)\varphi^\nu \equiv -a_u\,U^\nu - a_\theta\vartheta^\nu - a_\phi\varphi^\nu \tag{2.1}$$

The norm of the acceleration four-vector is a Lorentz scalar and determines Liénards relativistic power formula

$$P = -\frac{2e^2}{3c^3}a^\nu a_\nu = \frac{2e^2}{3c^3}\left[a_u^2 + a_\theta^2 + a_\phi^2\right] \tag{2.2}$$

Functional forms of individual components may also be investigated. For this write components of the acceleration four-vector

$$a^\nu = [\gamma^4\boldsymbol{\beta}\cdot\boldsymbol{a},\, \gamma^2\boldsymbol{a} + \gamma^4(\boldsymbol{\beta}\cdot\boldsymbol{a})\boldsymbol{\beta}] = (\gamma^3\boldsymbol{\beta}\cdot\boldsymbol{a})\beta^\nu + [0, \gamma^2\boldsymbol{a}] \tag{2.3}$$

Terms on the far right indicate that only the space components of the four-vectors $(\vec{U}, \vec{\vartheta}, \vec{\varphi})$ are required to determine components of $\vec{a}$. First, identify the space components from

$$U^\nu = (U^o, \boldsymbol{U}) \qquad\qquad \vartheta^\nu = (\vartheta^o, \boldsymbol{\vartheta}) \qquad\qquad \varphi^\nu = (\varphi^o, \boldsymbol{\varphi}) \tag{2.4}$$

Then write contributions to the power formula as

$$a_u = -\gamma^2 \boldsymbol{a} \cdot \boldsymbol{U} = \gamma^3(\boldsymbol{a} \cdot \boldsymbol{\beta}) - \frac{\gamma}{1 - \boldsymbol{\beta} \cdot \hat{\boldsymbol{n}}}(\boldsymbol{a} \cdot \hat{\boldsymbol{n}}) \tag{2.5a}$$

$$a_\theta = -\gamma^2 \boldsymbol{a} \cdot \boldsymbol{\vartheta} = -\gamma^2(\boldsymbol{a} \cdot \hat{\boldsymbol{\theta}}) - \frac{\gamma^2}{1 - \boldsymbol{\beta} \cdot \hat{\boldsymbol{n}}}(\boldsymbol{a} \cdot \hat{\boldsymbol{n}}) \cdot (\boldsymbol{\beta} \cdot \hat{\boldsymbol{\theta}}) \tag{2.5b}$$

$$a_\phi = -\gamma^2 \boldsymbol{a} \cdot \boldsymbol{\varphi} = -\gamma^2(\boldsymbol{a} \cdot \hat{\boldsymbol{\phi}}) - \frac{\gamma^2}{1 - \boldsymbol{\beta} \cdot \hat{\boldsymbol{n}}}(\boldsymbol{a} \cdot \hat{\boldsymbol{n}}) \cdot (\boldsymbol{\beta} \cdot \hat{\boldsymbol{\phi}}) \tag{2.5c}$$

In terms of three vectors, the power formula is

$$P = \frac{2e^2}{3c^3}\gamma^4 \left[(\boldsymbol{a} \cdot \boldsymbol{U})^2 + (\boldsymbol{a} \cdot \boldsymbol{\vartheta})^2 + (\boldsymbol{a} \cdot \boldsymbol{\varphi})^2\right] \tag{2.6}$$

Inserting the functional forms of each term will then verify the “Cartesian” form of the power formula

$$P = \frac{2e^2}{3c^3}\left[\gamma^4 a^2 + \gamma^6(\boldsymbol{a} \cdot \boldsymbol{\beta})^2\right] \tag{2.7}$$

Each of (2.6) and (2.7) present their own methodology for determination of the power, and both easily reduce to the Larmor formula. Even more interesting though, is the ability to write a formula for a^2 in terms of the spacelike components of each spherical basis vector:

$$a^2 = \left[(\boldsymbol{a} \cdot \boldsymbol{U})^2 + (\boldsymbol{a} \cdot \boldsymbol{\vartheta})^2 + (\boldsymbol{a} \cdot \boldsymbol{\varphi})^2 - (\boldsymbol{a} \cdot \gamma\boldsymbol{\beta})^2\right] \tag{2.8}$$

This suggests the expansion

$$\begin{aligned} \boldsymbol{a} &= (\boldsymbol{a} \cdot \boldsymbol{U})\boldsymbol{U} + (\boldsymbol{a} \cdot \boldsymbol{\vartheta})\boldsymbol{\vartheta} + (\boldsymbol{a} \cdot \boldsymbol{\varphi})\boldsymbol{\varphi} - \gamma^2(\boldsymbol{a} \cdot \boldsymbol{\beta})\boldsymbol{\beta} \\ &= (\boldsymbol{a} \cdot \hat{\boldsymbol{n}})\hat{\boldsymbol{n}} + (\boldsymbol{a} \cdot \hat{\boldsymbol{\theta}})\hat{\boldsymbol{\theta}} + (\boldsymbol{a} \cdot \hat{\boldsymbol{\phi}})\hat{\boldsymbol{\phi}} \end{aligned} \tag{2.9}$$

2.2 Velocity Field Strength Tensor

In Minkowski space, each of 16 possible non-zero components of a second rank tensor are associated with two directions. A summary of these directions in the spherical and radiation bases are indicated in Figure 3.

Now suppose the velocity portion of the electrons' field strength tensor is to be written in the spherical basis. Components of this tensor in the Cartesian basis are typically written:

$$F_v^{\mu\nu} = \frac{e}{\rho^2}\left[U^\mu \beta^\nu - U^\nu \beta^\mu\right] \tag{2.10}$$

But a proper representation of $F_v^{\mu\nu}$ will include the basis vectors

$$\widehat{F}_v = \eta\left[U^\mu \beta^\nu - U^\nu \beta^\mu\right] \vec{\mathrm{e}}_\mu \otimes \vec{\mathrm{e}}_\nu = \eta[\vec{U}\vec{\beta} - \vec{\beta}\vec{U}] \tag{2.11}$$

	$\vec{\beta}$	$\vec{U}$	$\vec{\vartheta}$	$\vec{\varphi}$
$\vec{\beta}$	×	×	×	×
$\vec{U}$	×	×	×	×
$\vec{\vartheta}$	×	×	×	×
$\vec{\varphi}$	×	×	×	×

	$\hat{\boldsymbol{t}}$	$\hat{\boldsymbol{n}}$	$\hat{\boldsymbol{\theta}}$	$\hat{\boldsymbol{\phi}}$
$\hat{\boldsymbol{t}}$	×	×	×	×
$\hat{\boldsymbol{n}}$	×	×	×	×
$\hat{\boldsymbol{\theta}}$	×	×	×	×
$\hat{\boldsymbol{\phi}}$	×	×	×	×

Figure 3: *Sixteen possible bi-derectional components of an arbitrary second rank tensor* $\widehat{T}$ *in the spherical basis and the radiation basis.*

Accrodingly, $\widehat{F}_v$ has only two non-zero components in the spherical basis proportional to the dilatation η. In a matrix format components are

$$F_v^{\mu\nu} \longrightarrow \begin{bmatrix} 0 & -\eta & 0 & 0 \\ \eta & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{2.12}$$

It's also possible to determine $\widehat{F}_v$ using the gradient operator (1.31) applied to the Lienard-Wiéchert potentials. Write $\vec{\nabla}\vec{A} = \partial^\mu A^\nu \cdot \vec{\mathrm{e}}_\mu \otimes \vec{\mathrm{e}}_\nu$ with components determined from

$$\partial^\mu A^\nu = \left[\beta^\mu \frac{\partial}{\partial c\tau} - U^\mu \frac{\partial}{\partial \rho} - \vartheta^\mu \frac{1}{R}\frac{\partial}{\partial \theta} - \varphi^\mu \frac{1}{R\sin\theta}\frac{\partial}{\partial \phi}\right] \cdot \frac{e}{\rho}\beta^\nu \tag{2.13}$$

Since β^ν is constant, only the derivative with respect to ρ is needed to generate $\widehat{F}$ from the anti-symmetric combination

$$\widehat{F} = [\partial^\mu A^\nu - \partial^\nu A^\mu]\,\vec{\mathrm{e}}_\mu \otimes \vec{\mathrm{e}}_\nu = \eta[\vec{U}\vec{\beta} - \vec{\beta}\vec{U}] \tag{2.14}$$

To re-write $\widehat{F}$ in the radiation basis, replace $\vec{\beta}$ and $\vec{U}$ as they are written in equation (1.56).

2.3 Acceleration Field Stength Tensor

Unfortunately, difficulties arise when using the gradient operator in (2.13) to calculate particle acceleration fields. The source of the problem can be traced to the calculation of the spacelike unit vector $\vec{U}$ which generalizes during particle accelerations

$$\vec{\nabla}\rho = -\vec{U} + \frac{a^\lambda R_\lambda}{\rho}\vec{R} \tag{2.15}$$

2.3.1 Spherical Basis:

A more appropriate choice is to work with vacuum gauge acceleration potentials

$$\vec{A}_a = -a^\lambda R_\lambda \vec{A} \qquad \text{where} \qquad \vec{A} = \frac{eR^\nu}{\rho^2}\vec{\mathrm{e}}_\nu \tag{2.16}$$

This set of potentials are not associated with velocity fields—and instead, yield the acceleration field strength tensor directly from

$$F_a^{\mu\nu} = \partial^\mu A_a^\nu - \partial^\nu A_a^\mu \tag{2.17}$$

Applying the chain rule determines

$$\partial^\mu A_a^\nu = -\partial^\mu[a^\lambda R_\lambda]\cdot A^\nu - a^\lambda R_\lambda \cdot \partial^\mu A^\nu \tag{2.18}$$

The first term is more difficult to calculate,

$$\begin{aligned}\partial^\mu[a^\lambda R_\lambda] &= \left[\beta^\mu\partial_\tau - U^\mu\partial_\rho - \frac{1}{R}\vartheta^\mu\partial_\theta - \frac{1}{R\sin\theta}\varphi^\mu\partial_\phi\right]\rho\cdot a^\lambda U_\lambda \\ &= -U^\mu a^\lambda U_\lambda + \beta^\mu\rho\frac{\partial a^\lambda}{\partial c\tau}U_\lambda - \rho U^\mu\frac{\partial a^\lambda}{\partial\rho}U_\lambda - \frac{\rho}{R}\vartheta^\mu a^\lambda\frac{\partial U_\lambda}{\partial\theta} - \frac{\rho}{R\sin\theta}\varphi^\mu a^\lambda\frac{\partial U_\lambda}{\partial\phi}\end{aligned} \tag{2.19}$$

The second line cleans up nicely, by combining derivatives of the four-acceleration and using derivatives of U^ν in equation (1.8). This leaves

$$\partial^\mu[a^\lambda R_\lambda] = -U^\mu a^\lambda U_\lambda + R^\mu\frac{\partial a^\lambda}{\partial c\tau}U_\lambda - \vartheta^\mu a^\lambda\vartheta_\lambda - \varphi^\mu a^\lambda\varphi_\lambda \tag{2.20}$$

Gathering terms together in (2.17) leads to several cancellations while generating components of the acceleration field strength tensor in the form

$$F_a^{\mu\nu} = a_\theta[\vartheta^\mu, A^\nu] + a_\phi[\varphi^\mu, A^\nu] \tag{2.21}$$

This is a useful result as each term on the right can be associated with one of two possible polarizations of emitted radiation. Additionally, (2.21) indicates the usefulness of vacuum gauge potentials when writing the theory of particle acceleration fields in retarded Minkowski space. Vector and tensor symbols can also be appended to (2.21) leading to the concise formulation

$$\widehat{F}_a = a_\theta[\vec{\vartheta}, \vec{A}] + a_\phi[\vec{\varphi}, \vec{A}] \tag{2.22}$$

In terms of spherical basis unit vectors, independent polarizations of $\widehat{F}_a$ are,

$$\widehat{F}_a(\theta) = a_\theta\frac{e}{\rho}[\vec{\vartheta}, \vec{\beta}+\vec{U}] \qquad\qquad \widehat{F}_a(\phi) = a_\phi\frac{e}{\rho}[\vec{\varphi}, \vec{\beta}+\vec{U}] \tag{2.23}$$

and lead directly to the anti- block diagonal matrix representation:

$$F_a^{\mu\nu} \longrightarrow \frac{e}{\rho}\begin{bmatrix} 0 & 0 & -a_\theta & -a_\phi \\ 0 & 0 & -a_\theta & -a_\phi \\ a_\theta & a_\theta & 0 & 0 \\ a_\phi & a_\phi & 0 & 0 \end{bmatrix} \tag{2.24}$$

2.3.2 Radiation Basis

Although the spherical basis is quite successful in its covariant description of the electron, the basis vectors are complicated, making it difficult to get useful information from tensor components. This problem can be eliminated for the acceleration fields by transforming to the radiation basis. First write vacuum gauge potentials as

$$\vec{A} = V\hat{\boldsymbol{t}} + V\hat{\boldsymbol{n}} \qquad \text{where} \qquad V \equiv \frac{eR}{\rho^2} \tag{2.25}$$

From equation (1.56) write

$$\vec{\theta} = \left[\frac{\boldsymbol{\beta}\cdot\hat{\boldsymbol{\theta}}}{1-\hat{\boldsymbol{n}}\cdot\boldsymbol{\beta}}\right]\hat{\boldsymbol{t}} + \left[\frac{\boldsymbol{\beta}\cdot\hat{\boldsymbol{\theta}}}{1-\hat{\boldsymbol{n}}\cdot\boldsymbol{\beta}}\right]\hat{\boldsymbol{n}} + \hat{\boldsymbol{\theta}} \tag{2.26a}$$

$$\vec{\phi} = \left[\frac{\boldsymbol{\beta}\cdot\hat{\boldsymbol{\phi}}}{1-\hat{\boldsymbol{n}}\cdot\boldsymbol{\beta}}\right]\hat{\boldsymbol{t}} + \left[\frac{\boldsymbol{\beta}\cdot\hat{\boldsymbol{\phi}}}{1-\hat{\boldsymbol{n}}\cdot\boldsymbol{\beta}}\right]\hat{\boldsymbol{n}} + \hat{\boldsymbol{\phi}} \tag{2.26b}$$

Inserting (2.25) and (2.26) into (2.22) elicits many cancellations leaving the radiation basis formulation of the acceleration field strength:

$$\widehat{F_a} = a_\theta V[\hat{\boldsymbol{\theta}}, \hat{\boldsymbol{t}}] + a_\theta V[\hat{\boldsymbol{\theta}}, \hat{\boldsymbol{n}}] + a_\phi V[\hat{\boldsymbol{\phi}}, \hat{\boldsymbol{t}}] + a_\phi V[\hat{\boldsymbol{\phi}}, \hat{\boldsymbol{n}}] \tag{2.27}$$

Components of $\widehat{F_a}$ in the new basis are

$$F_a^{\mu\nu} \longrightarrow V \begin{bmatrix} 0 & 0 & -a_\theta & -a_\phi \\ 0 & 0 & -a_\theta & -a_\phi \\ a_\theta & a_\theta & 0 & 0 \\ a_\phi & a_\phi & 0 & 0 \end{bmatrix} \tag{2.28}$$

This is almost identical to (2.24) except for the multiplier in front.

2.3.3 Analysis of Acceleration Fields

Using (2.27) and (2.28) it is possible to construct individual components of transverse electric and magnetic acceleration fields[4]:

$$\mathbf{E}_\theta = V a_\theta \hat{\boldsymbol{\theta}} \qquad\qquad \mathbf{B}_\phi = V a_\theta \hat{\boldsymbol{\phi}} \tag{2.30a}$$

$$\mathbf{E}_\phi = V a_\phi \hat{\boldsymbol{\phi}} \qquad\qquad \mathbf{B}_\theta = -V a_\phi \hat{\boldsymbol{\theta}} \tag{2.30b}$$

[4]Electric and magnetic field vectors written here can be reconciled with fields given in Cartesian coordinates by making the connection

$$a_\theta \hat{\boldsymbol{\theta}} + a_\phi \hat{\boldsymbol{\phi}} = \frac{\gamma^2}{1-\hat{\boldsymbol{n}}\cdot\boldsymbol{\beta}}\left[\hat{\boldsymbol{n}} \times (\hat{\boldsymbol{n}} - \boldsymbol{\beta}) \times \boldsymbol{a}\right] \tag{2.29}$$

Polarization vectors on the left are assumed to have no time component and if re-labeled as four-vectors ϵ_θ^ν and ϵ_ϕ^ν then the Lorentz condition is $\epsilon_\theta^\nu p_\nu = \epsilon_\phi^\nu p_\nu = 0$.

Each linear polarization state can be associated with energy flux specified by the Poynting vector $\mathbf{S}_a = \mathbf{S}_1 + \mathbf{S}_2$, with individual flux components written precisely in terms of components of vacuum gauge velocity potentials:

$$\mathbf{S}_1 = \frac{c}{4\pi}\mathbf{E}_\theta \times \mathbf{B}_\phi = \frac{1}{4\pi c^3} a_\theta^2 V\boldsymbol{A} \tag{2.31a}$$

$$\mathbf{S}_2 = \frac{c}{4\pi}\mathbf{E}_\phi \times \mathbf{B}_\theta = \frac{1}{4\pi c^3} a_\phi^2 V\boldsymbol{A} \tag{2.31b}$$

Energy density in the fields takes on a simple form in the Radiation basis. It can be extracted directly from the energy flux and determines energy density associated with each field polarization.

$$\mathcal{U}_a = \frac{1}{8\pi}\left(\mathrm{E}_a^2 + \mathrm{B}_a^2\right) = \frac{1}{4\pi}V^2(a_\theta^2 + a_\phi^2) \qquad\qquad \mathbf{S}_a = c\,\mathcal{U}_a\hat{\boldsymbol{n}} \tag{2.32}$$

Likewise, radiated power per unit solid angle for each polarization follows by writing

$$\frac{dP}{d\Omega} = (1 - \hat{\boldsymbol{n}}\cdot\boldsymbol{\beta})R^2\left(\mathbf{S}_1 + \mathbf{S}_2\right)\cdot\hat{\boldsymbol{n}} = \frac{dP_1}{d\Omega} + \frac{dP_2}{d\Omega} \tag{2.33}$$

2.4 Circular Polarization States

It is useful to construct fields with circular polarization by including an appropropriate phase between individual components

$$\mathbf{E}_a = \mathbf{E}_\theta + i\mathbf{E}_\phi \tag{2.34a}$$

$$\mathbf{B}_a = \mathbf{B}_\phi + i\mathbf{B}_\theta \tag{2.34b}$$

Unfortunately, electric and magnetic fields along directions $\hat{\boldsymbol{\theta}}$ and $\hat{\boldsymbol{\phi}}$ do not generally have the same magnitude as they might for an optics experiment. A way around this problem is to arrange portions of left- and right- circular polarization states from the definitions

$$\mathbf{E}_R = \frac{1}{\sqrt{2}}V(a_\theta + a_\phi)\boldsymbol{\epsilon} \qquad\qquad \mathbf{B}_R = -i\mathbf{E}_R \tag{2.35a}$$

$$\mathbf{E}_L = \frac{1}{\sqrt{2}}V(a_\theta - a_\phi)\boldsymbol{\epsilon}^* \qquad\qquad \mathbf{B}_L = i\mathbf{E}_L \tag{2.35b}$$

In these formulas, each of the fields are composed of electric and magnetic components from equation (2.30). Field directions are determined by complex polarization vectors $\boldsymbol{\epsilon}$ and $\boldsymbol{\epsilon}^*$ defined by

$$\boldsymbol{\epsilon} = \frac{1}{\sqrt{2}}\left(\hat{\boldsymbol{\theta}} + i\hat{\boldsymbol{\phi}}\right) \tag{2.36}$$

and are therefore eigenstates of the SO(3) photon helicity operator. The Poynting Vector is still $\mathbf{S}_a$ in this basis, but now left- and right- handed energy flux formulas (including factors of c) are:

$$\mathbf{S}_R = \frac{c}{4\pi}\mathbf{E}_R \times \mathbf{B}^*_R = \frac{1}{8\pi c^3}(a_\theta + a_\phi)^2 V\boldsymbol{A} \tag{2.37a}$$

$$\mathbf{S}_L = \frac{c}{4\pi}\mathbf{E}_L \times \mathbf{B}^*_L = \frac{1}{8\pi c^3}(a_\theta - a_\phi)^2 V\boldsymbol{A} \tag{2.37b}$$

Once again, left- and right-handed components of the energy density can be extracted directly from the energy flux yielding

$$\mathcal{U}_R = \frac{1}{8\pi}\left(|\mathbf{E}_R|^2 + |\mathbf{B}_R|^2\right) = \frac{1}{c}\mathbf{S}_R \cdot \hat{\boldsymbol{n}} = \frac{1}{8\pi}V^2(a_\theta + a_\phi)^2 \tag{2.38a}$$

$$\mathcal{U}_L = \frac{1}{8\pi}\left(|\mathbf{E}_L|^2 + |\mathbf{B}_L|^2\right) = \frac{1}{c}\mathbf{S}_L \cdot \hat{\boldsymbol{n}} = \frac{1}{8\pi}V^2(a_\theta - a_\phi)^2 \tag{2.38b}$$

Radiated power is still (2.33) except projected onto left- and right-handed components:

$$\frac{dP}{d\Omega} = (1 - \hat{\boldsymbol{n}} \cdot \boldsymbol{\beta})R^2\left(\mathbf{S}_L + \mathbf{S}_R\right) \cdot \hat{\boldsymbol{n}} = \frac{dP_L}{d\Omega} + \frac{dP_R}{d\Omega} \tag{2.39}$$

and therefore conserving angular momemtum at all times.

Example 5: Proof of Parity Symmetry: Unlike the weak nuclear force, the electromagnetic interaction respects parity symmetry. For an accelerating classical electron, this requires that left-handed and right-handed spin-1 photons be emitted with equal probabilities—therefore conserving angular momentum. To prove parity symmetry, it is necessary to show that

$$\oint \frac{dP_L}{d\Omega}d\Omega = \oint \frac{dP_R}{d\Omega}d\Omega \tag{2.40}$$

which follows if

$$I = \oint (1 - \boldsymbol{\beta} \cdot \hat{\boldsymbol{n}}) \cdot a_\theta a_\phi \, d\Omega = 0 \tag{2.41}$$

From equation (2.5) it appears to be a complicated integral, but simplifies considerably if we choose $\boldsymbol{a} \perp \hat{\boldsymbol{n}}$ or $\boldsymbol{\beta} \parallel \hat{\boldsymbol{n}}$. The relevant integral reduces to

$$I = \oint (1 - \boldsymbol{\beta} \cdot \hat{\boldsymbol{n}})(\boldsymbol{a} \cdot \hat{\boldsymbol{\theta}})(\boldsymbol{a} \cdot \hat{\boldsymbol{\phi}})d\Omega = 0 \qquad Q.E.D. \tag{2.42}$$

Example 5: Electric Dipole Radiation: A simple application of the formalism developed here is the solution to the problem of electric dipole radiation. Assume an electron is oscillating at frequency ω about the origin of a coordinate system with displacement along the z-axis. The position, velocity, and acceleration of the particle are:

$$\boldsymbol{z}(\tau) = z_o \sin\omega\tau \qquad \boldsymbol{v}(\tau) = \omega z_o \cos\omega\tau \qquad \boldsymbol{a}(\tau) = -\omega^2 z_o \sin\omega\tau \tag{2.43}$$

Evaluation of a_θ and a_ϕ in equation (2.5) indicate $a_\phi = 0$ and

$$a_\theta = -\omega^2 z_o \sin\theta \sin\omega\tau + \frac{\omega^2 z_o^2}{c} \cos\theta \sin\theta \cos\omega\tau \sin\omega\tau \tag{2.44}$$

Keeping only the term linear in the small diplacement z_o, the average radiated energy flux calculated from equation (2.31) is

$$\langle \mathbf{S}_1 \rangle = \left[\frac{e^2\omega^4 z_o^2}{8\pi c^3}\right] \boldsymbol{\cdot} \frac{\sin^2\theta}{r^2}\hat{\boldsymbol{n}} \tag{2.45}$$

But since $a_\phi = 0$, equations (2.37) and (2.38) further indicate that the radiated flux and resulting energy density are composed of equal amounts of left-handed and right-handed photons. Moreover, these photons are all radiated at the frequency of the oscillating electron.

Example 6: Bremsstrahlung: Braking radiation occurs when a beam of high energy electrons strike a metal target. An analysis of this problem assumes the beam velocity is $\boldsymbol{\beta} = \beta\hat{\boldsymbol{z}}$ with a rapid negative acceleration $\boldsymbol{a} = -a\hat{\boldsymbol{z}}$ at impact. For this problem, once again $a_\phi = 0$ requiring radiation to be composed of equal portions of right- and left-handed flux. Additionally,

$$a_\theta = -\frac{\gamma^2 a \sin\theta}{1-\beta\cos\theta} \qquad \mathbf{S}_{R/L} = \frac{e^2a^2}{8\pi c^3} \cdot \frac{\sin^2\theta}{R^2(1-\beta\cos\theta)^6}\hat{\boldsymbol{n}} \tag{2.46}$$

leading to power formulas per unit solid angle

$$\frac{dP_L}{d\Omega} = \frac{dP_R}{d\Omega} = \frac{e^2a^2}{8\pi c^3} \cdot \frac{\sin^2\theta}{(1-\beta\cos\theta)^5} \tag{2.47}$$

Example 7: Synchrotron Radiation: A beam of high energy electrons in circular motion emits synchrotron radiation. For this problem particle velocity and acceleration may be written $\boldsymbol{\beta} = \beta\hat{\boldsymbol{z}}$ and $\boldsymbol{a} = a\hat{\boldsymbol{x}}$ and satisfy $\boldsymbol{a}\boldsymbol{\cdot}\boldsymbol{\beta} = 0$. Unlike the previous two examples, both a_θ and a_ϕ make non-zero contributions to radiated power adding complexity to the problem:

$$a_\theta = -\gamma^2 a \cos\theta\cos\phi + \frac{\gamma^2 a\beta}{1-\beta\cos\theta}\sin^2\theta\cos\phi \qquad a_\phi = \gamma^2 a \sin\phi \tag{2.48}$$

The essential solution for radiated power per unit solid angle is

$$\frac{dP}{d\Omega} = \frac{e^2 a^2}{4\pi c^3}\left[\frac{(1-\beta\cos\theta)^2 - (1-\beta^2)\sin^2\theta\cos^2\phi}{(1-\beta\cos\theta)^5}\right] \tag{2.49}$$

In terms of left and right components, it is reasonable to write

$$\frac{dP_{\scriptscriptstyle R}}{d\Omega} = \frac{1}{2}\left[\frac{dP}{d\Omega} + \frac{dP'}{d\Omega}\right] \qquad\qquad \frac{dP_{\scriptscriptstyle L}}{d\Omega} = \frac{1}{2}\left[\frac{dP}{d\Omega} - \frac{dP'}{d\Omega}\right] \tag{2.50}$$

where

$$\frac{dP'}{d\Omega} = \frac{e^2 a^2}{4\pi c^3}\cdot\frac{(\beta-\cos\theta)\sin\phi\cos\phi}{(1-\beta\cos\theta)^4} \tag{2.51}$$

While the power per unit solid angle shows variations with angle the total radiated power is still one-half left-handed and one-half right-handed.

2.5 Symmetric Stress Tensor

The Lagrangian for the electromagnetic field is

$$\mathcal{L}_{em} = -\frac{1}{16\pi}F_{\mu\nu}F^{\mu\nu} - \frac{1}{c}j_e^\nu A_\nu \tag{2.52}$$

Source Maxwell equations follow by application of Lagrange's equations, but components of the symmetrized stress tensor can also be generated from $\mathcal{L}_{em}$ by writing

$$\Theta^{\mu\nu} = \frac{1}{4\pi}\left[F^\mu{}_\lambda F^{\lambda\nu} + \frac{1}{4}\mathrm{g}^{\mu\nu}F_{\alpha\beta}F^{\alpha\beta}\right] \tag{2.53}$$

For the most general (accelerated) motions of the electron, the stress tensor is a complicated equation[5]. It can be organized by writing

$$\Theta^{\mu\nu} = \Theta_1^{\mu\nu} + \Theta_2^{\mu\nu} + \Theta_3^{\mu\nu} \tag{2.54}$$

where individual components of the tensor fall off like ρ^{-4}, ρ^{-3}, and ρ^{-2}, respectively.

Stress Tensor in the Spherical Basis: In the absence of particle accelerations, equation (2.10) can be inserted into (2.53) which determines

$$\Theta_1^{\mu\nu} = \frac{1}{4\pi}\eta^2\left[\beta^\mu\beta^\nu - U^\mu U^\nu - \tfrac{1}{2}\mathrm{g}^{\mu\nu}\right] \tag{2.55}$$

[5] See Rohrlich's book, *Classical Charged Particles*, chapter 5 which writes the most general stress tensor in a cartesian basis.

But components of the metric tensor may be written

$$\mathrm{g}^{\mu\nu} = \beta^\mu\beta^\nu - U^\mu U^\nu - \vartheta^\mu\vartheta^\nu - \varphi^\mu\varphi^\nu \tag{2.56}$$

This leaves $\Theta_1^{\mu\nu}$ diagonalized in the spherical basis, and only a function of η^2

$$\Theta_1^{\mu\nu} = \frac{1}{8\pi}\eta^2\left[\beta^\mu\beta^\nu - U^\mu U^\nu + \vartheta^\mu\vartheta^\nu + \varphi^\mu\varphi^\nu\right] \longrightarrow \frac{1}{8\pi}\begin{bmatrix} \eta^2 & 0 & 0 & 0 \\ 0 & -\eta^2 & 0 & 0 \\ 0 & 0 & \eta^2 & 0 \\ 0 & 0 & 0 & \eta^2 \end{bmatrix} \tag{2.57}$$

Particle accelerations can be included by inserting $F^{\mu\nu}_{total} = F^{\mu\nu}_v + F^{\mu\nu}_a$ into equation (2.53). In the spherical basis, the matrix representation for $\Theta^{\mu\nu}$ is therefore

$$\Theta^{\mu\nu} \rightarrow \frac{1}{4\pi}\eta^2 \begin{bmatrix} \frac{1}{2} + \rho^2(a_\theta^2 + a_\phi^2) & +\rho^2(a_\theta^2 + a_\phi^2) & -\rho a_\theta & -\rho a_\phi \\ +\rho^2(a_\theta^2 + a_\phi^2) & -\frac{1}{2} + \rho^2(a_\theta^2 + a_\phi^2) & -\rho a_\theta & -\rho a_\phi \\ -\rho a_\theta & -\rho a_\theta & \frac{1}{2} & 0 \\ -\rho a_\phi & -\rho a_\phi & 0 & \frac{1}{2} \end{bmatrix} \tag{2.58}$$

This representation should be compared with its cumbersome and lengthy Cartesian counterpart found in many textbooks. The matrix above is not only concise, but each of the two components linear and quadratic in the quantities a_θ and a_ϕ is easily discernable by inspection. More precisely, one can write

$$\Theta_2^{\mu\nu} = -\frac{1}{4\pi\rho}\begin{bmatrix} 0 & \mathcal{F}_a \\ \mathcal{F}_a^{\mathrm{T}} & 0 \end{bmatrix} \qquad \Theta_3^{\mu\nu} = \frac{1}{4\pi}\begin{bmatrix} \mathcal{F}_a\mathcal{F}_a^{\mathrm{T}} & 0 \\ 0 & 0 \end{bmatrix} \tag{2.59}$$

A Adding Polarization States to Jackson

For the most general motions of the electron, transversely polarized radiation is composed of a wide range of frequencies. It is practical to follow Jackson, section 14.5, with the added benefit that the field in our problem is neatly divided into independent polarization states indicated by (2.30). Labeling the polarization states by s, define the quantity

$$Y_s(t) = \sqrt{\frac{1}{c^3}}\, RAa_s(t) \tag{A.1}$$

The power radiated per unit solid angle then follows as

$$\frac{dP}{d\Omega} = \frac{1}{4\pi}\sum_{s=\theta,\phi}|Y_s(t)|^2 \tag{A.2}$$

Fourier integrals of components $Y_s(t)$ can be determined by transforming the components $a_s(t)$

$$a_s(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} a_s(t) e^{i\omega t} dt \tag{A.3}$$

In general, the quantities $a_s(\omega)$ will have both real and complex components. According to Jackson, equation 14.57, the total energy radiated per unit solid angle is

$$\frac{dW}{d\Omega} = \frac{R^2 A^2}{4\pi c^3} \sum_{s=\theta,\phi} \int_{-\infty}^{\infty} a_s^*(\omega) a_s(\omega) d\omega = \frac{1}{4\pi} \sum_{s=\theta,\phi} \int_{-\infty}^{\infty} Y_s^*(\omega) Y_s(\omega) d\omega \tag{A.4}$$

Before proceeding any further, it will be necessary to re-formulate this equation to integrate over only positive frequencies. This is possible because $a_s(t)$ is a real quantity implying $a_s(-\omega) = a_s^*(\omega)$, and therefore

$$\frac{dW}{d\Omega} = \frac{1}{2\pi} \sum_{s=\theta,\phi} \int_{0}^{\infty} Y_s^*(\omega) Y_s(\omega) d\omega \tag{A.5}$$

B Complete Set of Commutators

Unit vectors associated with the spherical basis can be combined to form six commutators:

$$\mathcal{H}_1^{\mu\nu} = [\beta^\mu,\, U^\nu] \qquad \mathcal{H}_2^{\mu\nu} = [\beta^\mu,\, \vartheta^\nu] \qquad \mathcal{H}_3^{\mu\nu} = [\beta^\mu,\, \varphi^\nu] \tag{B.1a}$$
$$\mathcal{H}_4^{\mu\nu} = [U^\mu,\, \vartheta^\nu] \qquad \mathcal{H}_5^{\mu\nu} = [U^\mu,\, \varphi^\nu] \qquad \mathcal{H}_6^{\mu\nu} = [\vartheta^\mu,\, \varphi^\nu] \tag{B.1b}$$

An interesting exercise to use the commutators to write the total field strength tensor for the classical electron

$$\begin{aligned} F_{total}^{\mu\nu} &= \eta\,[U^\mu, \beta^\nu] - a_\theta \eta [R^\mu,\, \vartheta^\nu] - a_\phi \eta [R^\mu,\, \varphi^\nu] \\ &= \eta \mathcal{H}_1^{\mu\nu} - a_\theta \frac{e}{\rho}\,[\mathcal{H}_2^{\mu\nu} + \mathcal{H}_4^{\mu\nu}] + a_\phi \frac{e}{\rho}\,[\mathcal{H}_3^{\mu\nu} + \mathcal{H}_5^{\mu\nu}] \end{aligned} \tag{B.2}$$

As the final commutator $\mathcal{H}_6^{\mu\nu}$ is not needed in this expansion, this encourages an investigation which determines

$$\eta \mathcal{H}_6^{\mu\nu} = \eta[\vartheta^\mu,\, \varphi^\nu] \equiv F_D^{\mu\nu} \tag{B.3}$$

where $F_D^{\mu\nu}$ is the dual to the velocity portion of the field strength tensor. This promotes the idea that classical electromagnetism spans the space of anti-symmetric rank-2 tensors.